Lesson 12 Watering regularly

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1. Overview

In Lesson 11, we uploaded code to turn this kit into an alarm device that can set the alarm time according to needs. I believe you have already implemented this feature according to the course and would like to get more gameplay. This kit can not only be used as an alarm device, but also as a watering device.

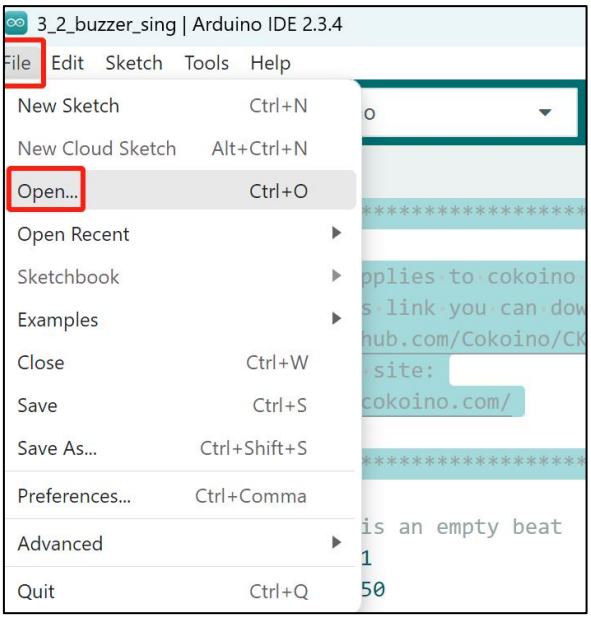
In this Lesson, we will upload code to the Arduino UNO board to make this kit a watering device. And by uploading different codes, the functions of this watering device will also be different. For example, when uploading the code "9_1_Watering_by_unting_time", the device can set a cycle countdown for watering. When uploading the code "9_2_Watering_by_day", the device can set daily scheduled watering. When uploading the code "9_3_Watering_by_week", the device can set weekly scheduled watering.

2. Select and open the code

The code used in this lesson is placed in this folder：“ E:\CKK0021-main\Tutorial\sketches”

Double-click the Arduino IDE shortcut on the desktop to open it

Click on the Arduino IDE interface toolbar"File" --- "open"



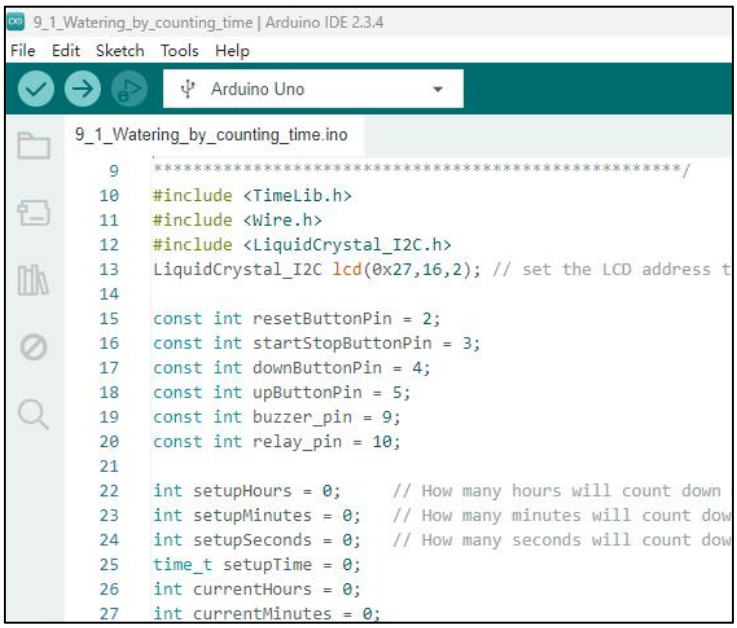
2.1 Select Countdown Watering Function Code

Select the code in the folder named 9_1_Watering_by_counting_time：

9_1_Watering_by_counting_time

E:\CKK0021-main\Tutorial\sketches\9_1_Watering_by_counting_time,Then click "open".

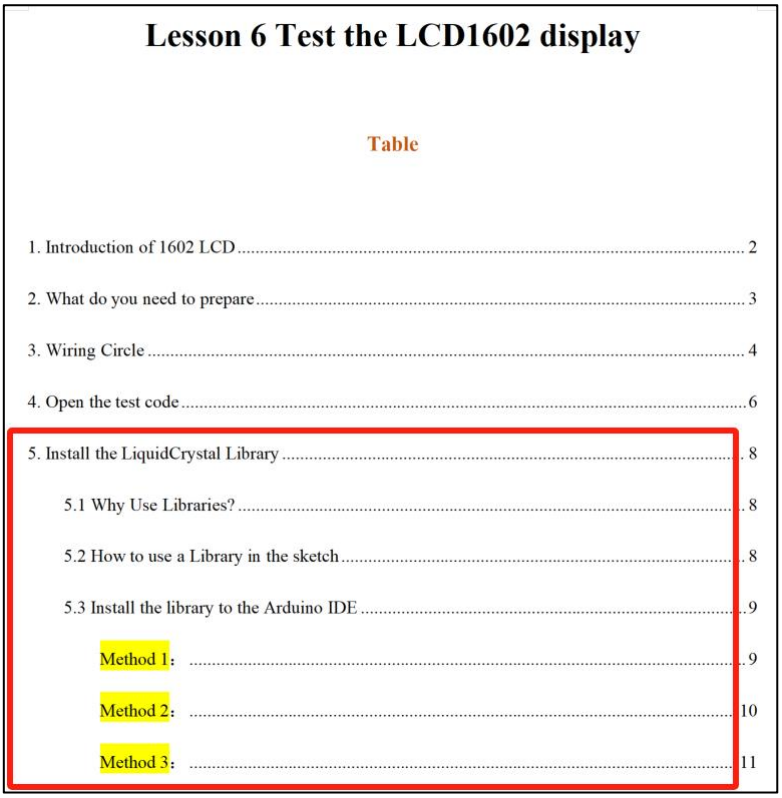
The code interface after opening is as follows



Install the Time library

To install the Time library,please refer to the Item 5 of **Lesson 6** for library installation methods

"5. Install the Liquid Crystal Library"



Lesson 6 Test the LCD1602 display

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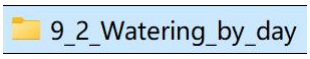
2.2 Select the daily scheduled watering function code

Select the code in the folder named 9_2_Watering_by_day：



E:\CKK0021-main\Tutorial\sketches\9_2_Watering_by_day,Then click "open".

The code interface after opening is as follows

9_2_Watering_by_day | Arduino IDE 2.3.4

File Edit Sketch Tools Help

Select Board

9_2_Watering_by_day.ino

```
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2); // set the LCD add
#include <DS1302.h>
#include <stdio.h>

const int kCePin   = 8;  // Chip Enable
const int kIoPin   = 6;  // Input/Output
const int kSclkPin = 7;  // Serial Clock
// 初始化DS1302
DS1302 rtc(kCePin, kIoPin, kSclkPin);

// 按键引脚
const int resetButtonPin = 2;
const int startStopButtonPin = 3;
const int downButtonPin = 4;
const int upButtonPin = 5;
const int buzzerpin = 9;
```

2.3 Select the weekly scheduled watering function code

Select the code in the folder named 9_3_Watering_by_week：

9_3_Watering_by_week

E:\CKK0021-main\Tutorial\sketches\ 9_3_Watering_by_week,Then click "open".

The code interface after opening is as follows

9_3_Watering_by_week | Arduino IDE 2.3.4

File Edit Sketch Tools Help

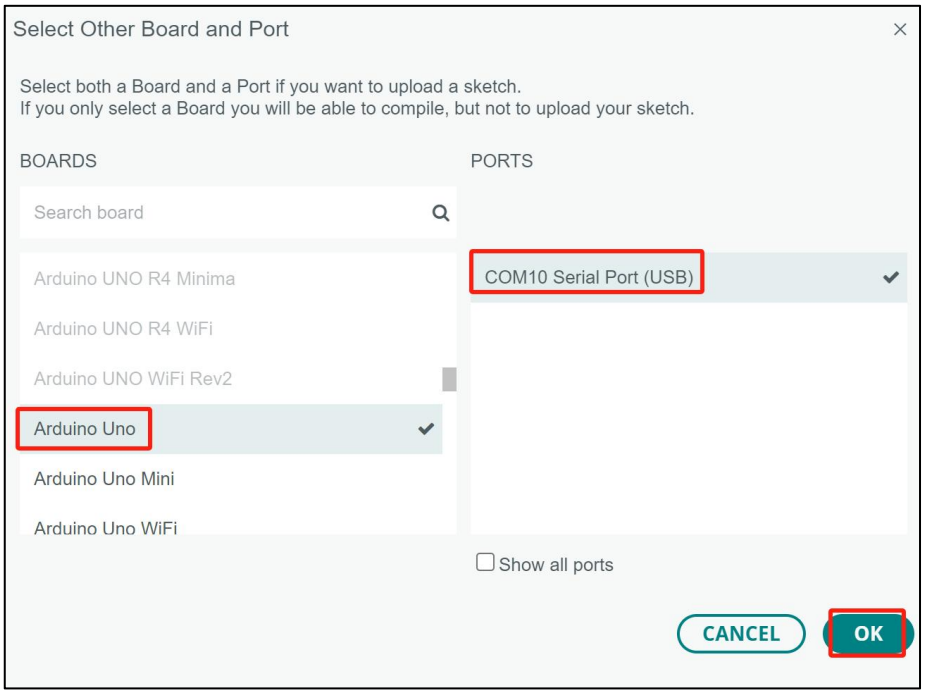
Select Board

9_3_Watering_by_week.ino

```
 * Company web site:
 * http://www.cokoino.com/

*****************************************************/
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2); // set the LCD address
#include <DS1302.h>
#include <stdio.h>

const int kCePin   = 6;  // Chip Enable
const int kIoPin   = 7;  // Input/Output
const int kSclkPin = 8;  // Serial Clock
// 初始化DS1302
DS1302 rtc(kCePin, kIoPin, kSclkPin);

// 按键引脚
const int resetButtonPin = 2;
const int startStopButtonPin = 3;
const int downButtonPin = 4;
const int upButtonPin = 5;
const int buzzerpin = 9;
const int relayPin = 10;
```

3. Upload the code

After selecting and opening the code, connect the computer and Arduino UNO board using an A to B USB cable.Select the board "Arduino UNO"and Port "COM10" (COM port is commonly known as an input output port for a device normally PC which enables communication between Arduino and PC. You can check your arduino com number in device manager, the com port of our arduino board is recognized as COM10 in this tutorial)



Click compile button, successfully compiled the code will display "Done compiling"

Click upload button, successfully uploading the code will display "Done uploading". When code is uploaded successfully, the program starts to run.

4. Build the watering system

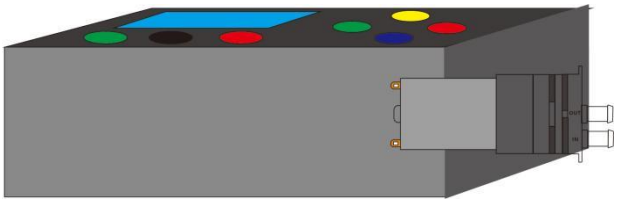
4.1 Connect the 4/7 black soft rubber hose with the tee fitting and the drip irrigation nozzle.

We have assembled this alarm clock and watering device according to Lesson 10, and now we can see that this kit also includes 3 adjustable drip irrigation nozzle, 3 tee fitting, 2 plugs, and a 5-meter-long 4/7 black hose. Using these things, along with the already assembled alarm clock and watering device, we will build a watering system.

Of course, potted plants or plants need to be prepared by yourself, and you also need to prepare a container filled with water.

Considering the power issue of the water pump, it is recommended that there should not be more than 3 pots or plants in the watering system, so this kit only comes with 3 adjustable sprinkler heads. This course builds a watering system and waters three plants at regular intervals. This is just a reference, you can also choose one or two plants to build a watering system.

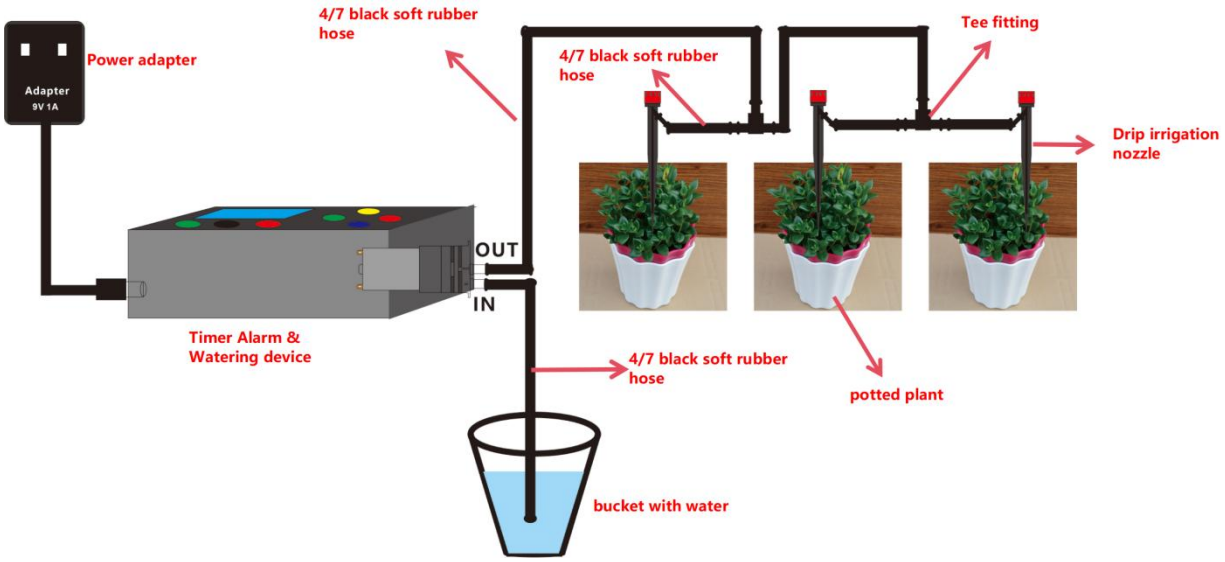
The required items are listed in the table below:

Components	Quantity	Picture	Remark
Timer Alarm & Watering device	1		You need to assemble it yourself according to Lesson 10
Tee fitting	2		
4/7 black soft rubber hose(5m)	1		Need to cut it yourself,you can cut the required length according to your actual situation
Drip irrigation nozzle	3		
Potted plants or plants	3		You need to prepare it yourself

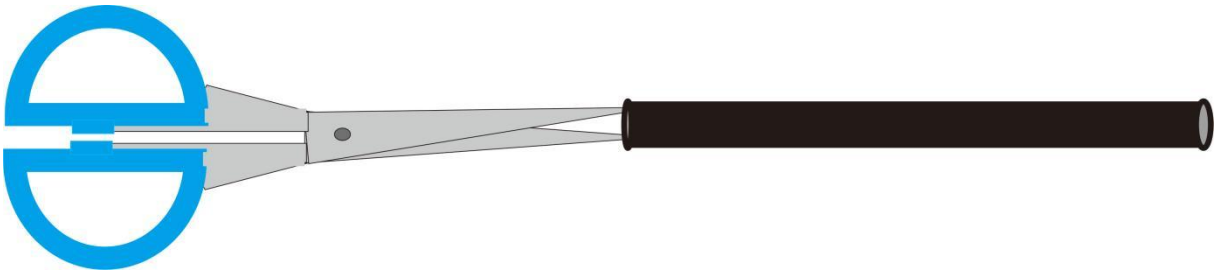
Bucket or other container (filled with water)	1		

As shown in the figure below, we use a 4/7 black soft rubber hose with one end connected to the IN port of the water pump and the other end inserted into the water bucket. Use a 4/7 black soft rubber hose, with one end connected to the OUT port of the water pump and the other end connected to a Tee fitting.

Use 4/7 black soft rubber hose to connect the Tee fitting and Drip irrigation nozzle. (Please cut the length of the 4/7 black soft rubber hose according to your own needs)



Note: The inner diameter of the 4/7 black soft rubber hose is smaller than the outer diameter of the Tee fitting to ensure sealing after connection and prevent water leakage. But it is quite difficult to connect, requiring a lot of effort to fit the 4/7 black soft rubber hose into the Tee fitting interface. If you feel like you can't fit it in, you can choose some conical tools, such as scissors, insert the 4/7 black soft rubber hose with scissors, and then rotate the scissors to enlarge the inner diameter of the opening of the 4/7 black soft rubber hose.



Finally, connect the Power adapter to the Timer Alarm&Watering device, turn the switch on the UNO expansion board to ON, and the Timer Alarm&Watering device will start up.

4.2 How to use the plug

Function of the plug: When one end of the 4/7 black soft rubber hose needs to be closed, insert the end cap into the 4/7 micro-tube.

Plug：A plug consists of two parts: the plug cap and the end cap.

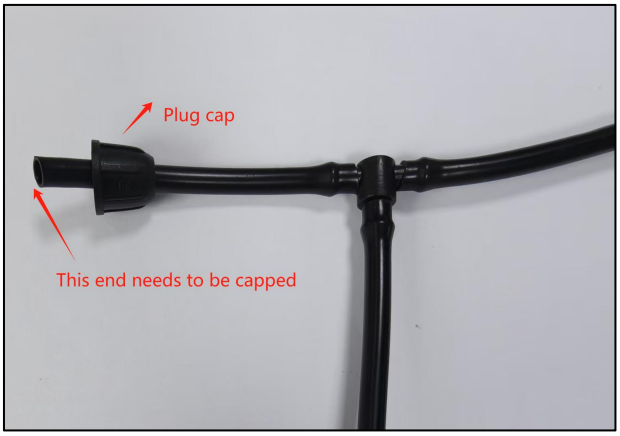
The picture is as follows：

Plug hat

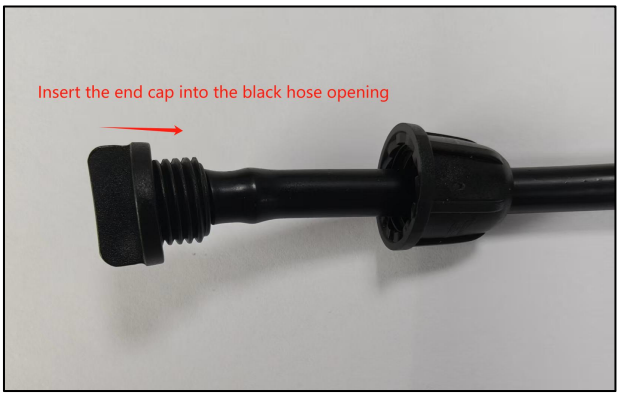
End cap

Usage Method：

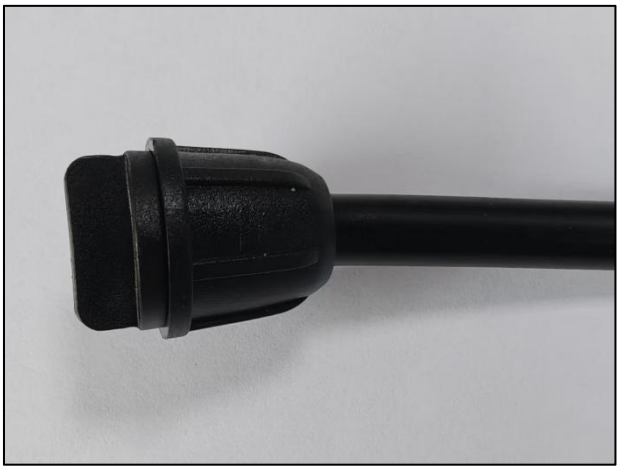
A，Pass the end of the 4/7 black hose that needs to be blocked through the plug cap.



B，Insert the end cap into the end of the 4/7 black hose that needs to be blocked.

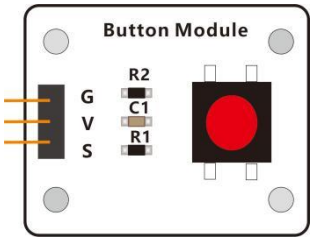


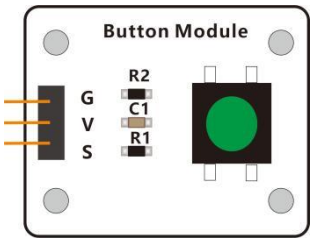
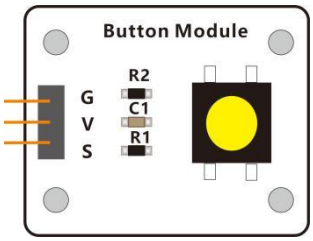
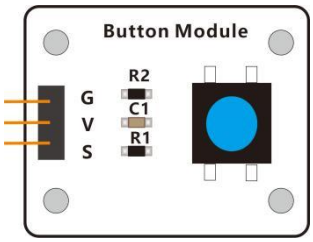
C，Turn the plug hat counterclockwise onto the end cap and tighten it securely.



5. Set timed watering time

Please refer to the table below for the functional definition of the button module：

Button Module	Picture	Name	Function
Red button		Reset button	1: Press and hold for 5 seconds to enter the settings mode 2: In the setting mode, press once to save the set time data and enter idle mode 3: In operating mode, press once to enter idle mode

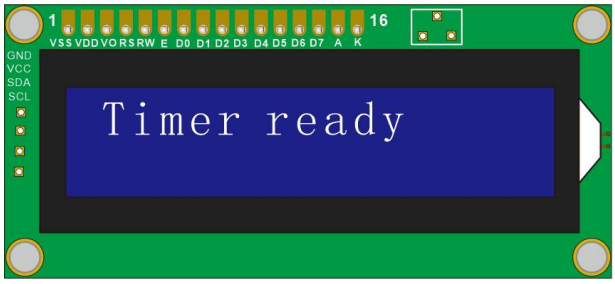
Green button		Start/Stop button	1: In idle mode, press once to enter run mode (start) 2: In operating mode, press once to enter idle mode (stop) 3: In setup mode, press this button to switch between Day of the week, hour, minute, and second
Yellow button		Up button	In the setting mode, it is used to add numbers for (Day of the week, hour, minute, and second) Adjustable numerical range: Day of the week (1-7), hours (00-23), minutes (00-59), seconds (00-59)
Blue button		Down button	In the setting mode, it is used to set the numerical subtraction for (Day of the week, hour, minute, and second) Adjustable numerical range:: Day of the week(7-1),hours (23-00) ,minutes (59-00) ,seconds (59-00)

5.1 Time setting for countdown watering

If you select and upload the countdown watering function code in steps 2 and 3, this Timer Alarm&Watering device can implement a cycle of countdown watering.

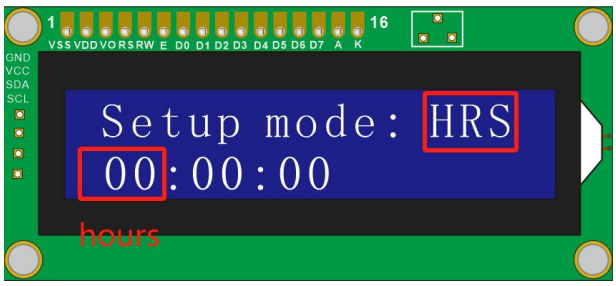
Note: Since the countdown function uses the timer function of Atemega328p to count, and uses the time function, it supports a maximum countdown of 9 hours and 6 minutes. Therefore, in code 9_1_Watering_by_counting_time, the maximum countdown time is set to 9 hours. When you set the time to 9, continue to press the yellow button with your hand, and the time value will not change to 10, but from 9 to 0. If you want to count down for a longer time, you can also modify the code yourself and use the time function combined with software counting.

After the Timer Alarm&Watering device is turned on, the green light lights up and the first line of the 1602 LCD screen displays "Timer ready"

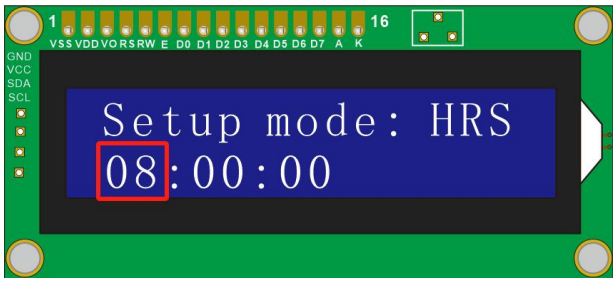


Next, use the button module to set the countdown watering time. Assuming we set this time to 8:10:55 .

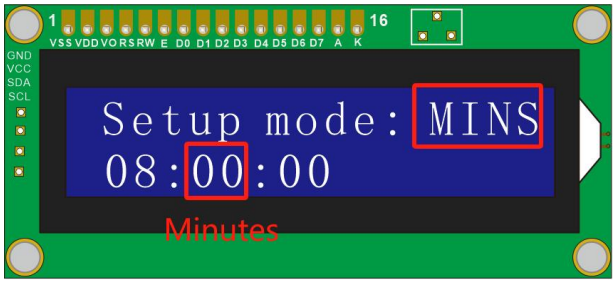
Firstly, we long press the Reset button for 5 seconds to enter the time setting mode, which defaults to setting the hour, as shown in the following figure:



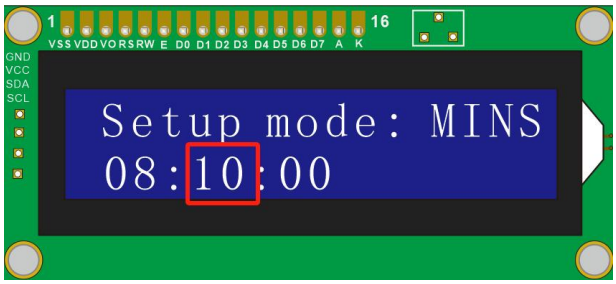
Next, press the yellow button 8 times with your hand, and the number on the second row of the 1602 LCD displaying the hour parameter will increase from 00 to 08.



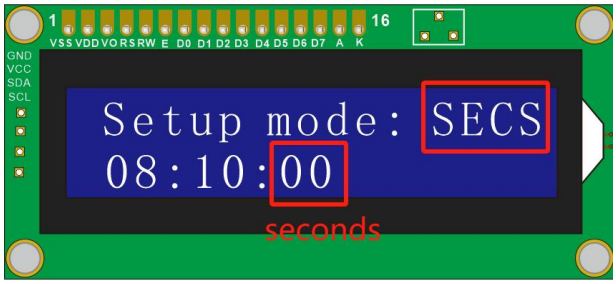
Then, we press the green button to change the setting mode from set hours to set minutes, as shown in the following figure:



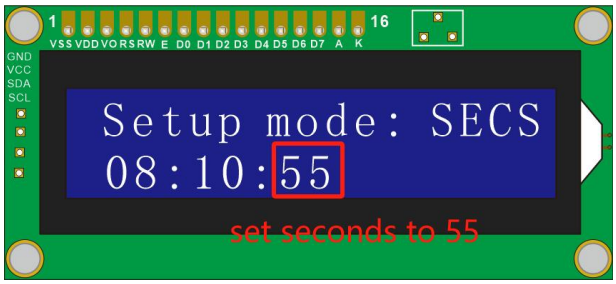
Next, press the yellow button 10 times with your hand, and the number in the second row of the 1602 LCD displaying the minute parameter will increase from 00 to 10.



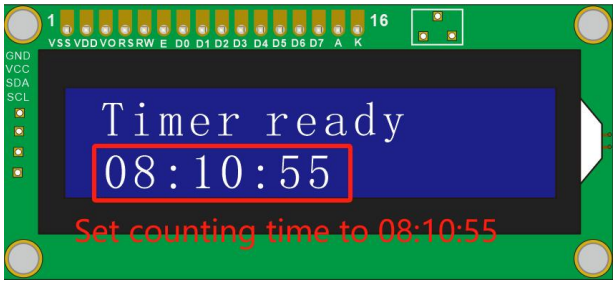
Then, we press the green button once to change the setting mode from set minutes to set seconds, as shown in the following figure:



Next, press the blue button 5 times with your hand, and the second line of the 1602 LCD will display the number from 00 to 55 for the second parameter.



Finally, we press the red button once to save the settings data and enter idle mode. At this point, the time parameter of 8:10:55 has been set.

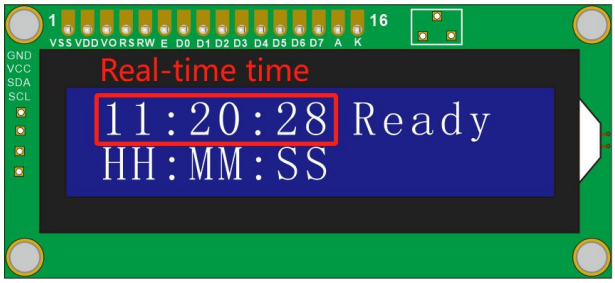


5.2 Time setting for daily watering

If you select and upload the daily scheduled watering function code in steps 2 and 3, this Timer Alarm&Watering device can achieve daily scheduled watering.

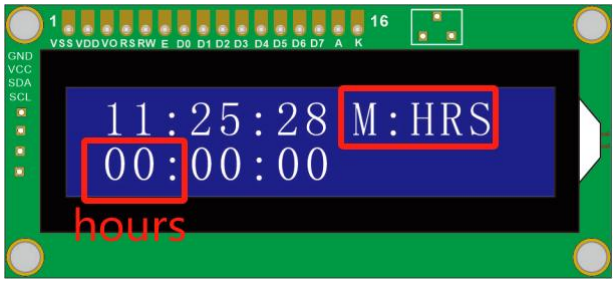
After the Timer Alarm&Watering device is turned on, the green light lights up, and the first line of the 1602 LCD screen displays the "Set Your Time:" string. Two seconds later, it displays the

real-time time and "Ready" string. The second line displays HH: MM: SS (time format, indicating that the time is to be set)

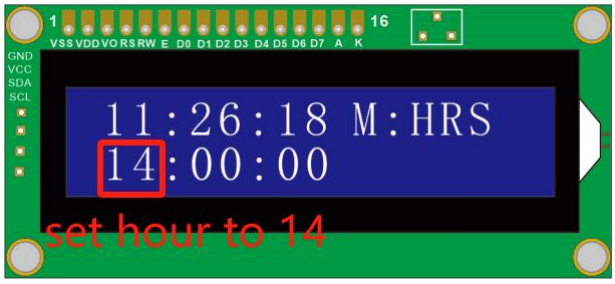


Next, use the button module to set the scheduled watering time for each day. Assuming we set this time to 14:50:00.

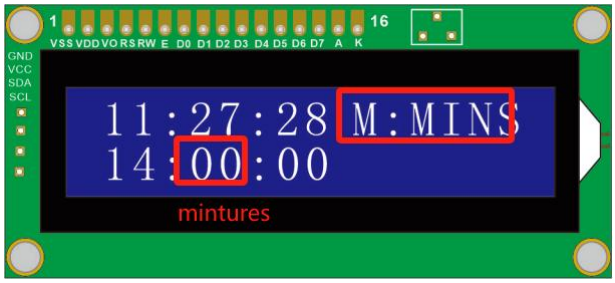
Firstly, we long press the Reset button for 5 seconds to enter the time setting mode, which defaults to setting the hour, as shown in the following figure:



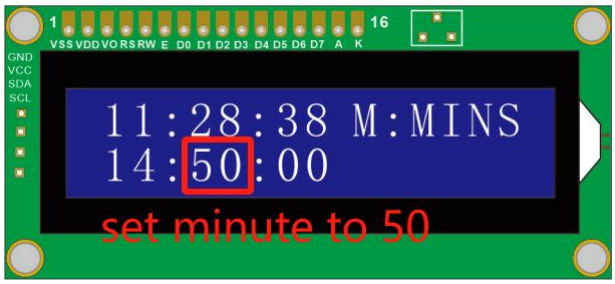
Next, press the yellow button 14 times with your hand, and the number for the hour parameter displayed on the second row of the 1602 LCD will increase from 00 to 14.



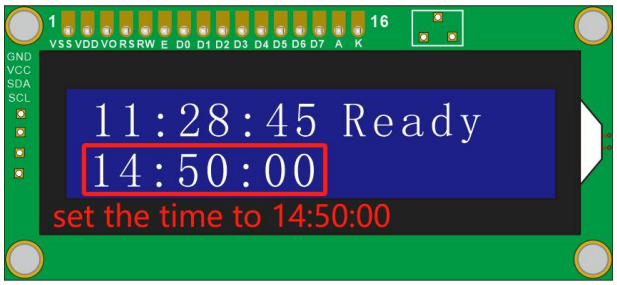
Then, we press the green button to change the setting mode from set hours to set minutes, as shown in the following figure



Press the blue button 10 times, and the number for the hour parameter displayed on the second row of the 1602 LCD will change from 00 to 59, and then decrease from 59 to 50.



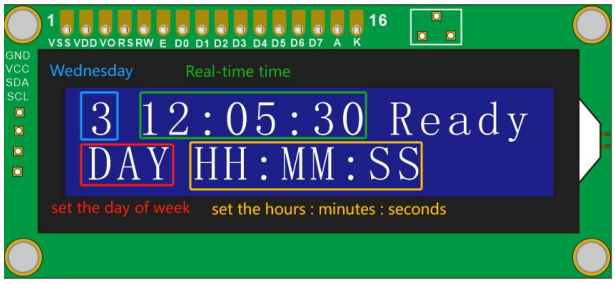
At this point, the time parameter of 14:50:00 has been set. We press the red button once to save the set data and enter idle mode.



5.3 Time setting for weekly watering schedule

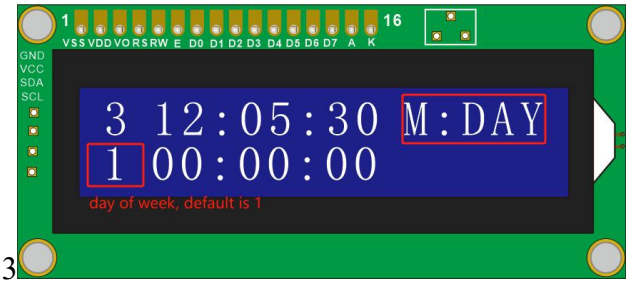
If you select and upload the weekly scheduled watering function code in steps 2 and 3, this Timer Alarm&Watering device can achieve weekly scheduled watering.

After the Timer Alarm&Watering device is turned on, the green light is on, and the first line of the 1602 LCD screen displays the "Set Your Time:" string. Two seconds later, it displays the day of week & real-time time and the "Ready" string. The second line displays “DAY HH: MM: SS” (time format, indicating time to be set)

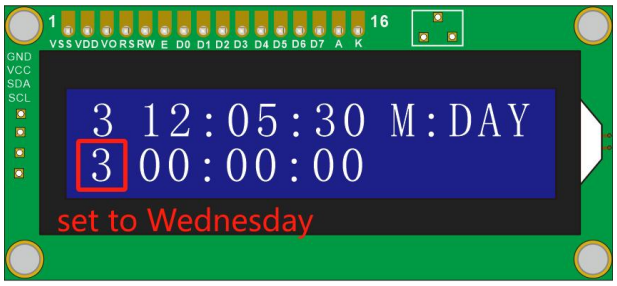


Next, use the button module to set the scheduled watering time for each week. Assuming we set this time as Wednesday at 14:50:05.

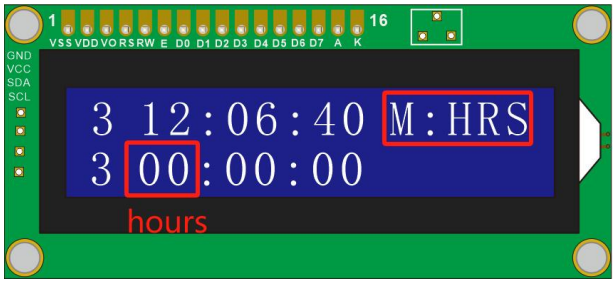
Firstly, we long press the Reset button for 5 seconds to enter the time setting mode. Upon entering, it defaults to setting the day of week and defaults to 1 (1 represents Monday, 2 represents Tuesday... 7 represents Sunday) As shown in the following figure:

3 

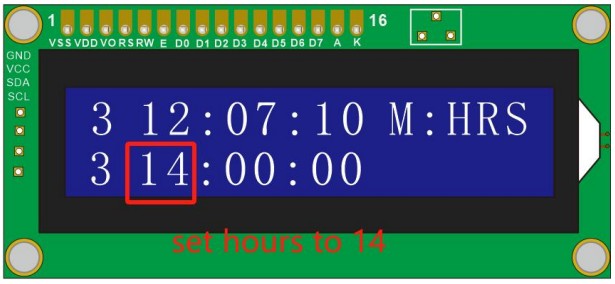
Next, press the yellow button twice with your hand, and the number of day of week displayed on the second row of the 1602 LCD will increase from 1 to 3.



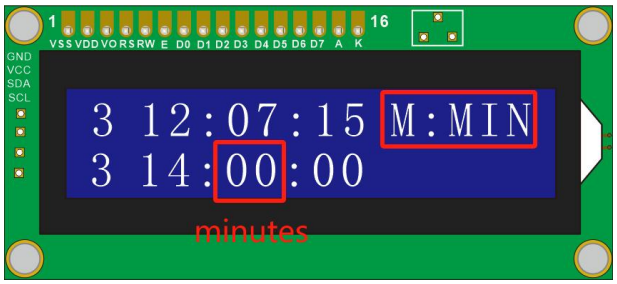
Next, press the green button once with your hand to change the setting mode from setting the day of the week to setting the hour, as shown in the following figure:



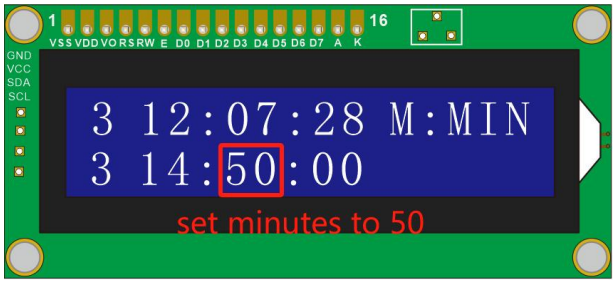
Then, press the blue button 10 times with your hand, and the value at hours will change from 00 to 23. Subtract 1 again until the value at hours becomes 14.



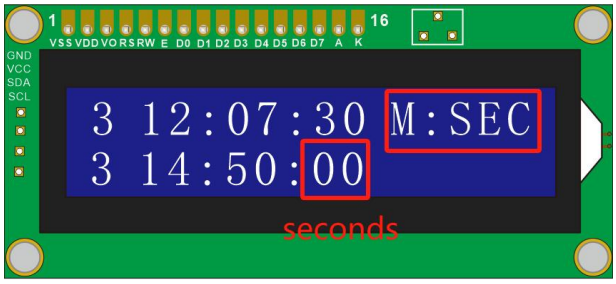
Then, we press the green button to change the setting mode from set hours to set minutes, as shown in the following figure：



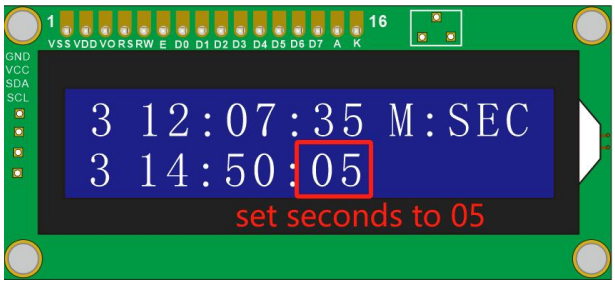
Next, press and hold the blue button 10 times until the value at minutes changes from 00 to 59, then subtract 1 each time until the value at minutes becomes 50.



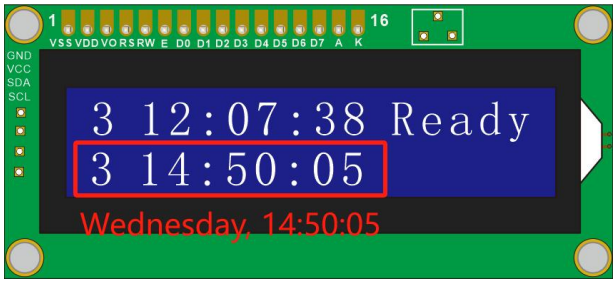
Then, we press the green button to change the setting mode from set minutes to set seconds, as shown in the following figure:



Next, press the yellow button 5 times with your hand, and the value at seconds will change from 00 to 01. Then, add 1 again until the value at seconds becomes 05.



At this point, the time parameter (Wednesday, 14:50:05) has been set. Press the red button with your hand to save the set parameter and enter idle mode.

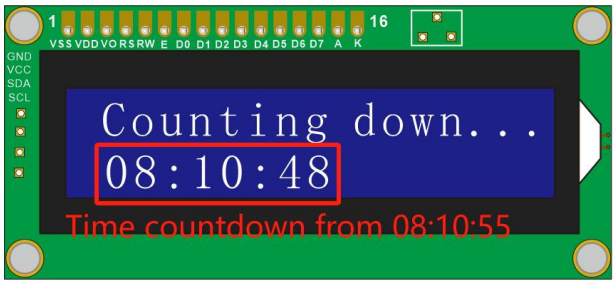


6. Run watering function

6.1 Countdown watering

Press the green (ST) button on the Timer Alarm&Watering device to enter the running mode.

The countdown starts from 08:10:55 and decreases gradually



When the countdown reaches 00:00:00, the buzzer sounds once, the green light goes out, and the red light lights up. With a beep of the relay, the water pump began to operate with a buzzing sound, and the first row of the 1602 LCD displayed "Watering". The sprinkler dripper inserted into the potted plant begins to spray water,

After 5 seconds, the water pump stops working, the red light goes out, and the green light comes on. Continuing with the next round of countdown watering mode, without changing the parameter settings, when the second round of countdown reaches 00:00:00, the watering function will be activated for another 5 seconds, and then enter the third round of countdown, repeating the cycle.

Of course, this is just a demonstration, you can set different times according to your own needs. Refer to step 5.1 of this course for the setting method

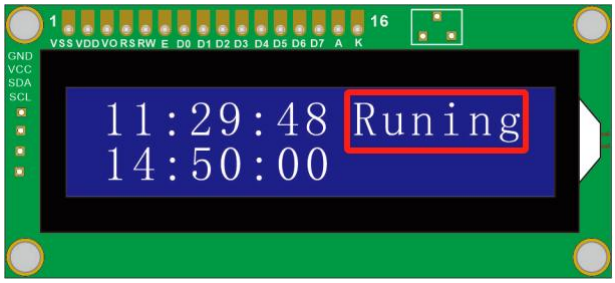
The watering time for the example code is 5 seconds, as shown in the following figure at line 306 of the code.

You can change this time according to your own needs.

```
301     case MODE_Watering:
302       buzzer_ring();
303       digitalWrite(LED_G,LOW);//The green LED is off
304       digitalWrite(LED_R,HIGH);//The red LED is on
305       digitalWrite(relay_pin,HIGH);//Relay closed, power supply to water pump
306       delay(5000);//The water pump works for 5 seconds, you can customize this time
307       digitalWrite(relay_pin,LOW);//Relay disconnected, water pump stops working
308       digitalWrite(LED_R,LOW);//The red LED is off
309       digitalWrite(LED_G,HIGH);////The green LED is on
310       currentMode = MODE_IDLE;
311       flag = true;
312       break;
313   }
```

6.2 Water the plants regularly every day

Press the green (ST) button on the Timer Alarm&Watering device to enter the running mode.



The system compares the real-time time with the set watering time. When the real-time time reaches the set watering time, the buzzer starts to work and you will hear a "beep-beep" sound for about two seconds. Then the buzzer stops working, the green light turns off and the red light turns on. With a "click" from the relay, the water pump starts to hum continuously, and "Waterin" is displayed at the end of the first line on the 1602LCD screen. The irrigation dripper inserted in the flower pot begins to spray water.

After 10 seconds, the water pump stops working, the red light turns off and the green light turns on again. The system then resumes the next cycle of the timed watering mode. Without changing the parameter settings, the watering function will be activated on time at 14:50:00 the next day.

Of course, this is just a demonstration. You can set different times according to your own needs. For the setting method, refer to Step 5.2 of this course.

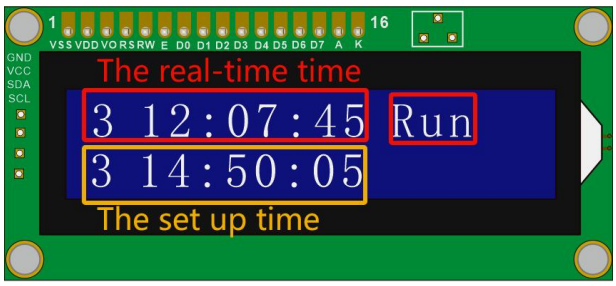
The watering duration in the sample code is set to 10 seconds, as shown in the figure below (Line 319 of the code). You can modify this duration as required.

9_2_Watering_by_day.ino

```
306        break;
307      case MODE_RUNNING:
308      //When the real-time time reaches the set dayly watering time, the running mode switc
309        if(t.hr == setupHours && t.min == setupMinutes && t.sec == setupSeconds)
310        {
311          currentMode = MODE_Watering;
312        }
313        break;
314      case MODE_Watering:
315        buzzer_ring();
316        digitalWrite(LED_G,LOW);//The green LED is off
317        digitalWrite(LED_R,HIGH);//The red LED is on
318        digitalWrite(relay_pin,HIGH);//Relay closed, power supply to water pump
319        delay(10000);//The water pump works for 10 seconds, you can customize this time
320        digitalWrite(relay_pin,LOW);//Relay disconnected, water pump stops working
321        digitalWrite(LED_R,LOW);//The red LED is off
322        digitalWrite(LED_G,HIGH);////The green LED is on
323        currentMode = MODE_IDLE;
324        flag = true;
325        break;
```

6.3 Timed Watering on a Weekly Basis

Press the green (ST) button on the Timer Alarm&Watering device to enter the running mode.



The system compares the real-time time with the set watering time. When the real-time time reaches the set watering time (Wednesday, 14:50:05), the green light turns off and the red light turns on. Then the buzzer starts to work, and you can hear a "beep-beep" sound for about two seconds before the buzzer stops working. The relay emits a "click" sound, the water pump starts to hum

continuously, and "Water" is displayed at the end of the first line on the 1602LCD screen. The irrigation dripper inserted in the potted plant starts to spray water.

After 30 seconds, the water pump stops working, the red light turns off, and the green light turns on. The system then proceeds to the next round of the timed watering mode. Without changing the parameter settings, the watering function will start on time again at 14:50:00 on Wednesday of the second week.

Of course, this is just a demonstration. You can set different times according to your own needs. For the setting method, refer to Step 5.3 of this course.

The watering duration in the sample code is 30 seconds, as shown in the figure below (Line 363 of the code). You can modify this duration according to your own requirements.

```
9_3_Watering_by_week.ino
362    digitalWrite(relayPin,HIGH);//Relay closed, power supply to water pump
363    delay(30000);//The water pump works for 30 seconds, you can customize this time
364    digitalWrite(relayPin,LOW);//Relay disconnected, water pump stops working
365    digitalWrite(LED_R,LOW);//The red LED is off
366    digitalWrite(LED_G,HIGH);//The green LED is on
367    currentMode = MODE_IDLE;
```

7. Troubleshooting

A. Code compilation failed

Please check whether the required libraries for the code have been installed in the Arduino IDE. For the library installation method, please refer to Item 5 of Lesson 6, as shown in the figure below:

5. Install the LiquidCrystal Library 8

5.1 Why Use Libraries? 8

5.2 How to use a Library in the sketch 8

5.3 Install the library to the Arduino IDE 9

Method 1: 9

Method 2: 10

Method 3: 11

B. The time displayed in the real-time area on the first line at the top-left corner of the 1602LCD does not match the actual time of your computer.

Please follow Item 6 of Lesson9 to re-upload the code "7_2_get_and_set_real-time_time" to the Arduino UNO board. After confirming that the time displayed in the real- time time area on the 1602LCD is correct, re-upload the code you need to the Arduino UNO board.

If the time is still incorrect, please first upload the code "7_1_set_time_to_ds1302" to the Arduino UNO board according to Item 5 of Lesson 9, then re-upload the code

"7_2_get_and_set_real-time_time" to the Arduino UNO board. After the upload is successful, the real-time time displayed on the LCD will be correct. At this point, you can continue to upload the required functional application code to the Arduino UNO board.

C. The time setting function cannot be accessed using the buttons, or the time cannot be adjusted.
Please check whether the connection between the button modules and the UNO expansion board is correct and secure without looseness.

D. When the watering device performs the watering function, the Drip irrigation nozzle does not discharge water.
Please check whether the knob on the drip irrigation nozzle is tightened. The knob is used to adjust the water spray volume of the drip irrigation nozzle. If it is fully tightened, no water will flow out; the looser it is, the greater the water flow. However, do not loosen it too much, as the knob will fall off the drip irrigation nozzle if it is overly loose.

8. Any questions and suggestions are welcome

THANK YOU for participating in this Lesson 12 experience!

If you find any errors, omissions or you have suggestions and/or questions about this lesson, please feel free to contact us:

cokoino@outlook.com

We will make every effort to make changes and correct errors as soon as feasibly possible and publish a revised version.

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